

for all $x \in R$. Every potent ring R is reduced: if $x \in R$ satisfies $x^2 = 0$, let $n = n(x) > 1$ be such that $x^n = x$. Then $x = x^n = x^2 x^{n-1} = 0$. Every reduced ring is a subdirect product of domains [3, 16], but it would breach the spirit of this paper to use this rather deep structural result.

Every reduced ring R satisfies the condition

$$xy = 0 \implies yx = 0, \quad (2)$$

for all $x \in R$. Indeed, if $xy = 0$, then $(yx)^2 = y(xy)x = 0$ and so $yx = 0$ by (1). Rings satisfying (2) are said to be *reversible* [9].

Reduced rings are neither defined nor characterized by identities. Thus even elementary proofs in reduced rings unavoidably use (1) and hence are not, strictly speaking, equational. However, it is straightforward to convert such proofs to equational ones in n -potent rings. For example, to prove directly that an n -potent ring ($n > 1$) is reversible, note that if $xy = 0$, then $yx = (yx)^n = y(xy)^{n-1}x = 0$.

An idempotent e (that is, an element satisfying $e^2 = e$) of a reduced ring R is central. This is well known and has a short, standard proof: check that $(ex - exe)^2 = 0$ and $(xe - exe)^2 = 0$, so R being reduced implies $ex - exe = 0$ and $xe - exe = 0$, hence $ex = exe = xe$. In fact, the same expressions occur in a mild generalization.

Lemma 2.1. *In reversible rings, idempotents are central.*

Proof. If e is an idempotent in a ring R , then for all $x \in R$, $e(x - ex) = 0$ and $(x - xe)e = 0$. If R is reversible, then (2) implies $(x - ex)e = xe - exe = 0$ and $e(x - xe) = ex - exe = 0$. Thus $xe = exe = ex$, that is, e is central. $\square$

In the reduced case, the proof of the following useful generalization is only a bit more involved than the classic proof and is based on the same idea.

Lemma 2.2. *Let R be a reduced ring. If $c \in R$ satisfies $c^2 = tc$ for some integer t , then c is central.*

Proof. Firstly, for all $x \in R$, $c^2 xc = tc \cdot xc = cx \cdot tc = cxc^2$. Thus $c[c, x]c = c(cx - xc)c = 0$. In particular, $(c[c, x])^2 = 0$ and $([c, x]c)^2 = 0$. Since R is reduced, (1) yields both $c[c, x] = 0$ and $[c, x]c = 0$. These imply, respectively,

$$xc[c, x] = 0 \quad \text{and} \quad (3)$$

$$[c, x]cx = 0. \quad (4)$$

Applying (2) to (3), we get

$$[c, x]xc = 0. \quad (5)$$

Subtracting (5) from (4), we obtain $[c, x]^2 = 0$. Since R is reduced, $[c, x] = 0$ for all $x \in R$, that is, $c \in Z(R)$. $\square$

Lemma 2.3. *Let R be a reduced ring and let $n > 1$ be an integer. If $c \in R$ satisfies $c^n = c$, then c^{n-1} is a central idempotent.*

Proof. If $n = 2$, then $c^{n-1} = c$, while if $n > 2$, then $c^{n-1}c^{n-1} = c^n c^{n-2} = cc^{n-2} = c^{n-1}$. In either case, c^{n-1} is an idempotent and so Lemma 2.2 applies. $\square$

We now specialize from reduced rings to n -potent rings, our real interest. The following result will turn out to be crucial, and can be viewed as improving Lemma 2.3 in the case of odd n .

Lemma 2.4. *Let $k \geq 1$ be an integer and let R be a $(2k + 1)$ -potent ring. Then $x^k \in Z(R)$ for all $x \in R$.*

Proof. Firstly, x^{2k} is a central idempotent by Lemma 2.3. Next we show $x^{3k} = x^k$. The claim is clear if $k = 1$, while if $k > 1$, then $x^{3k} = x^{2k+1}x^{k-1} = xx^{k-1} = x^k$. Finally, $(x^{2k} + x^k)^2 = (x^{2k})^2 + 2x^{3k} + x^{2k} = 2(x^{2k} + x^k)$. By Lemma 2.2 (with $c = x^{2k} + x^k$, $t = 2$), $x^{2k} + x^k \in Z(R)$ and so $x^k = (x^{2k} + x^k) - x^{2k} \in Z(R)$ for all $x \in R$, as claimed. $\square$

For n even, we have the following.

Lemma 2.5. *Let $n > 1$ be an even integer and let R be an n -potent ring. Then $2x = 0$ for all $x \in R$.*

Proof. For all $x \in R$, $-x = (-x)^n = x^n = x$. $\square$

For the remainder of this section, we establish commutativity theorems for n -potent rings for various n . We start with a classic, recalling that 2-potent rings are the same as Boolean rings.

Theorem 2.6. *Every 2-potent ring is commutative.*

Proof. This is the case $n = 2$ of Lemma 2.3. $\square$

As an alternative, here is the classic proof.

Alternative proof 1. Let R be 2-potent. For all $x, y \in R$, $x + y = (x + y)^2 = x^2 + xy + yx + y^2 = x + xy + yx + y$. Cancelling, we have $xy + yx = 0$, and so $xy = -yx = yx$ by Lemma 2.5. $\square$

We also present here the stunning proof of Brandenburg [6].

Alternative proof 2. For all x, y in a ring R , $xy - yx =$

$$= ((x + y)^2 - (x + y)) - (x^2 - x) - (y^2 - y) + ((yx)^2 - yx) - ((-yx)^2 - (-yx)).$$

If R is 2-potent, then the right hand side equals 0. $\square$

For a plethora of proofs and variations of our next result, see [8].

Theorem 2.7. *Every 3-potent ring is commutative.*

Proof. This is the case $k = 1$ of Lemma 2.4. $\square$

Alternative proof. Let R be 3-potent. By Lemma 2.3, x^2 is central for all $x \in R$, so $2x = 2x^3 = (x^2 + x)^2 - x^4 - x^2$ is central. Next,

$$x^2 + x = (x^2 + x)^3 = x^6 + 3x^5 + 3x^4 + x^3 = x^2 + 3x + 3x^2 + x = 2 \cdot 2(x^2 + x) \in Z(R).$$

Finally, $x = (x^2 + x) - x^2 \in Z(R)$ for all $x \in R$, that is, R is commutative. $\square$

Lemma 2.8. *Let k be an integer and let R be a ring in which $k(xy + yx) \in Z(R)$ for all $x, y \in R$. Then $kx^2 \in Z(R)$ for all $x \in R$.*

Proof. We have $kx^2 \cdot y + kxyx = x \cdot k(xy + yx) = k(xy + yx) \cdot x = kxyx + y \cdot kx^2$ and so the desired result follows from canceling $kxyx$. $\square$

Lemma 2.9. *If R is a ring in which $x^2 + x \in Z(R)$ for all $x \in R$, then R is commutative.*

Proof. For all $x, y \in R$, we have $(x + y)^2 + x + y \in Z(R)$, that is, $(x^2 + x) + (y^2 + y) + xy + yx \in Z(R)$. It follows that $xy + yx \in Z(R)$, and thus $x^2 \in Z(R)$ by Lemma 2.8 (with $k = 1$). Therefore $x = (x^2 + x) - x^2 \in Z(R)$ for all $x \in R$, i.e., R is commutative. $\square$

Lemma 2.10. *If n is a power of 2, then each binomial coefficient $\binom{n}{k}$ is even except for $k = 0$ and $k = n$.*

Proof. This is easily established by induction. $\square$